

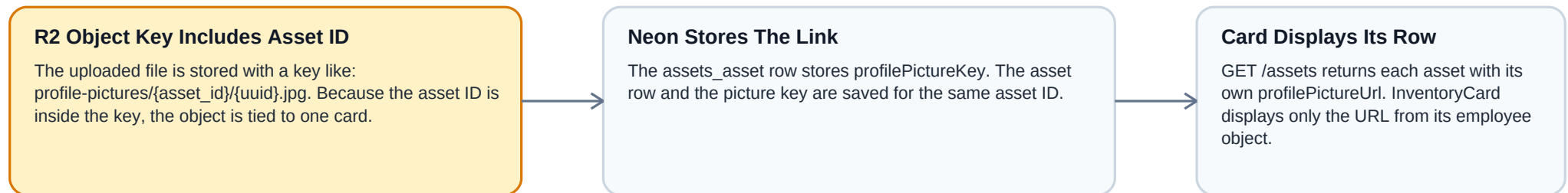
Profile Picture Upload Flow: React -> Django -> Cloudflare R2 -> Neon

Generated for the Invent hardware inventory project.

End-to-End Flow



How The Picture Stays On The Right Card



Storage Responsibilities

Cloudflare R2

Stores the raw image object. Example object key: profile-pictures/asset-123/abc123.jpg.

Neon PostgreSQL

Stores asset data and the R2 object key. It does not store the raw uploaded photo.

Frontend

Shows the picture URL returned by Django. It does not use R2 secret keys.

Relevant Code

1. frontend/src/components/EmployeeForm.tsx - avatar picker, image resize, data:image payload
2. backend/assets/views.py - save_profile_picture, asset create/update flow
3. backend/assets/r2_storage.py - decode data image, upload to R2, return signed/public URL
4. backend/assets/models.py - profilePictureKey stored on Asset, profilePictureUrl returned to frontend
5. backend/.env - R2_BUCKET_NAME, R2_ACCESS_KEY_ID, R2_SECRET_ACCESS_KEY, R2_ENDPOINT_URL

Important: R2 secret keys stay in backend/.env only. They are never sent to React.